Token Display System for Counselling Cell

Submitted To

Session: Jan-June

INSTITUTE OF TECHNOLOGY & MANAGEMENT

In Partial Fulfilment of the Degree of BACHELOR OF ENGINEERING IN ELECTRONICS & COMMUNICATION

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DECLARATION

We hereby declare that the Project entitled **"Token Display System for Counselling Cell"** is our own work conducted under the supervision of **Vineet Shrivastav, Department of Electronics & Communication Engineering at Institute of Technology and Management, Gwalior.**

We further declare that to the best of our knowledge this report does not contain any part of work that has been submitted for the award of any degree either in this university or in other
university without proper citation.

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CERTIFICATE

This is to certify that the work embodied in this project entitled**"TOKEN DISPLAY SYSTEM FOR COUNSELLING CELL"**being submitted by **Tanushka Mishra (0905EC231056), Raj Nandani Sharma (0905EC231042), and Stuti Pandey (0905EC231052)** in partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology (Electronics & Communication Engineering)** is a record of Bonafide piece of work, carried out under our supervision and guidance in the **Department of Electronics & Communication Engineering,Institute of Technology and Management, Gwalior (M.P.)**

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ACKNOWLEDGMENT

We express our heartfelt gratitude to our esteemed guide, **Mr. Vineet Shrivastava** and **Dr. Kapil Jain**, Head of the **Department of Electronics and Communication Engineering** , for their invaluable guidance, encouragement, and support throughout this project. Their deep knowledge, patience, and constructive feedback have been instrumental in helping us complete this work successfully.

We are also indebted to **Dr. S.S. Chauhan**, Dean Academics, Institute of Technology & Management for his valuable suggestions.

Our thanks extended to **Dr. Meenakshi Mazumdar**, Director, Institute of Technology & Management for his valuable suggestions.

We would like to thank to our institution and university for giving us such an opportunity to show our talent and present it to everyone.

We have to thank our parents for their love and support throughout my life. Thank you both for giving me strength to reach for the stars and chase my dreams.

We also extend our thanks to all the faculty members of the department for their continuous encouragement and assistance during various stages of the project. Their insights and suggestions enriched our learning experience and improved the quality of our work.

ABSTRACT

This project proposes a Token Display System for Counselling Cells in institutions, enabling efficient token calling and display. The system uses 4x4 keypads and I2C LCDs for each receptionist, interfaced with an Arduino Mega.

Tokens are shown individually on LCDs and collectively on a P10 LED scrolling display.The system ensures real-time updates, modular expansion, and easy usability, thereby reducing manual intervention.

Keywords:

Token Display, Arduino Mega, P10 LED Matrix, Keypad Interface, I2C LCD

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CHAPTER 1

INTRODUCTION

1.1 Background

In many institutions, managing queues in places like counselling cells is often chaotic, especially when multiple receptionists are handling student or visitor entries. Traditional systems are manual and prone to miscommunication, delays, and lack of synchronisation. An automated token display system using embedded electronics ensures better coordination, faster processing, and smoother communication between front desk staff and visitors.

1.2 Problem Statement

Manual queue management creates confusion and inefficiency. There is no centralised way for multiple receptionists to update the queue simultaneously. Hence, a system is needed where multiple users can input token numbers, and those tokens can be displayed centrally on a scrolling LED display.

1.3 Objectives

- Design a microcontroller-based system for managing token input from 4 users.
- Display real-time token numbers on a scrolling LED (P10).
- Use LCD displays for receptionist feedback.
- Ensure modularity and scalability of the system.

1.4 Scope of the Study

The project focuses on queue management within an institute. It is based on hardware and software integration using Arduino, keypads, LCDs, and P10 LED displays. Though limited to 4 inputs, it can be expanded further for hospitals, banks, and other public services.

Introduction

In today’s institutional environments, managing queues efficiently has become a critical need. Whether it is a counselling cell in a college, a hospital OPD, or a public service centre, an organised token system ensures smooth operations, better service flow, and reduced waiting time. Traditionally, token systems rely on manual call-outs or single-user digital solutions, which are neither scalable nor synchronised across multiple service desks.

To overcome these limitations, this project introduces a **microcontroller-based Token Display System** designed specifically for a counselling cell setup. The system enables up to four receptionists to operate independently, each using their own **4x4 matrix keypad** and **I2C LCD** to enter token numbers. These token numbers are then collectively displayed on a **P10 LED matrix display**, scrolling in real-time. This ensures centralised visibility while maintaining distributed input capability.

At the core of the system lies the **Arduino Mega 2560**, which acts as the central controller, receiving keypad inputs, managing LCD displays, and transmitting messages to the LED matrix. The use of **I2C communication** for LCDs and **SPI interface** for the P10 display allows efficient and simultaneous operation without performance compromise.

One of the key advantages of this system is **modularity**. It is designed in such a way that more reception units or display panels can be added with minor software and hardware tweaks. Moreover, with the popularity of open-source platforms like Arduino and the wide availability of components, this system is both **cost-effective** and **easy to implement**.

This project not only enhances operational efficiency but also demonstrates how **embedded systems** can solve real-life problems in institutional and public service domains. By integrating basic electronics, programming logic, and communication protocols, it showcases an excellent application of **Electronics & Communication Engineering** in developing smart solutions.

The Token Display System for Counselling Cell stands as a reliable, practical, and scalable solution for managing queues effectively. It serves as a foundation for future enhancements like mobile integration, wireless updates, or even voice- based announcements — paving the way for smart automation in day-to-day services.

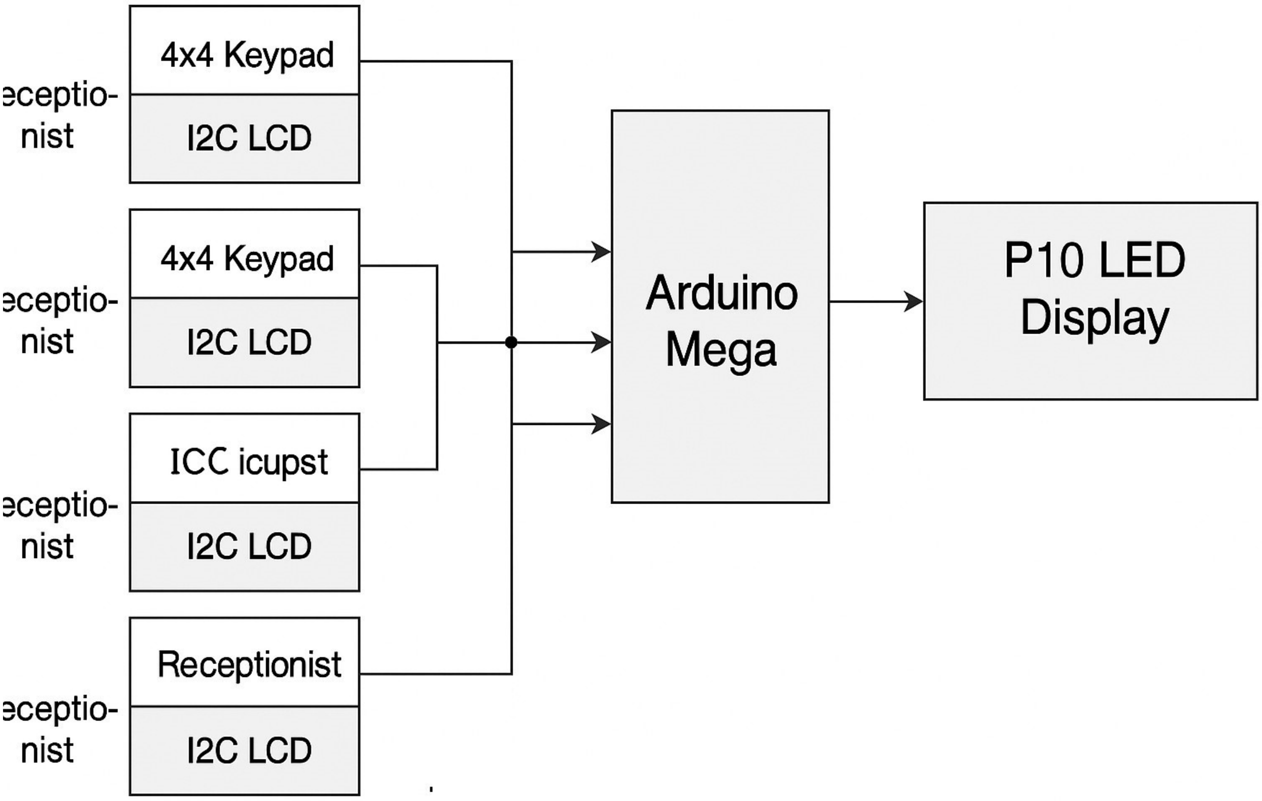


FIGURE 1.1. Flow chart of token system

CHAPTER 2
COMPONENTS AND MATERIALS REQUIRED

Components and Materials Required

Designing and implementing a token display system requires the integration of both hardware and software components. Each component plays a critical role in ensuring that the system operates smoothly, independently for each receptionist, while also coordinating centrally for display output.

This chapter provides a detailed overview of all the electronic components, microcontroller, display modules, input units, power supplies, and the software tools used for building the **Token Display System for Counselling Cell**.

2.1
Hardware Components

The following hardware components were used in the development of the system:

Arduino Mega 2560 (1x)

The Arduino Mega serves as the central processing unit. It is chosen for its large number of digital I/O pins and multiple serial communication ports, which are essential for handling four separate keypads and LCDs simultaneously along with the P10 LED module.

4x4 Matrix Keypads (4x)

Each receptionist is provided with a separate 4x4 matrix keypad to enter token numbers. The keypad consists of 16 tactile buttons arranged in a 4-row by 4-column configuration. The matrix allows scanning each key using only 8 Arduino pins.

I2C 16x2 LCD Displays (4x)

These displays are used to provide individual feedback to each receptionist. Using the I2C interface reduces wiring complexity by allowing all LCDs to share the same two communication lines (SDA and SCL), differentiated by unique I2C addresses.

P10 LED Display Modules (3x)

The 3 P10 modules (each 16x32 pixels) are connected in series to form a scrolling LED board. These are used to display all active tokens in real-time to the waiting visitors. Controlled by the MD_Parola and MD_MAX72XX libraries via SPI communication.

Power Supply (5V, 10A)

A 5V DC power supply with at least 10A current output is used to power all modules, especially the P10 panels which require high current.

Jumper Wires, Resistors, Breadboard, and Connectors

Used for internal wiring, prototyping, and connecting all components to the Arduino.

S.NO	Component	Quantity	Specifications
1	Arduino Mega 2560	1	54 digital I/O, 16 analog inputs
2	4x4 Matrix Keypad	4	Tactile button type, 16 keys
3	I2C 16x2 LCD Display	4	1602 LCD with I2C backpack
4	P10 LED Display	3	16x32 resolution,
5	Power Supply	1	5V DC, 10A
6	Jumper Wires	Multiple	Male-to-male, male-to-female

Table 2.1: List of Hardware

2.2 Software Requirements

The software environment consists of tools used for programming the Arduino and libraries to control the LCDs, keypads, and LED display.

1. Arduino IDE

An open-source platform used to write, compile, and upload code to the Arduino Mega. It supports C/C++ and allows easy integration of external libraries.

2. Required Libraries

- Keypad.h– to handle 4x4 matrix keypad inputs.
- LiquidCrystal_I2C.h– for I2C LCD communication.
- Wire.h– required for I2C protocol communication.
- MD_Parola.hand MD_MAX72XX.h– to control scrolling on P10 LED matrix.
- SPI.h– to enable SPI communication for the LED display.

SN0.1	Component	Arduino Mega Pins Used	Purpose
1	Keypad 1 (4x4)	22–29	Rows: 22–25, Columns: 26–29
2	Keypad 2 (4x4)	30–37	Rows: 30–33, Columns: 34–37
2	Keypad 2 (4x4)	38–45	Rows: 38–41, Columns: 42–45
3	Keypad 2 (4x4)	46–53	Rows: 46–49, Columns: 50–53
4	I2C LCDs (x4)	SDA: A4, SCL: A5	Shared I2C communication lines
5	P10 Display (3 Panels)	DIN: 9, CLK: 10, CS/STB: 11	SPI Communication (Serial Output)
6	P10 OE (Enable)	GND	Always kept LOW
7	LCD Power	5V, GND	Power Supply
8	P10 Power	5V (External), GND	High current power (10A recommended)

- **Table 2.2** Arduino Pin Configuration

Proper pin allocation ensures no overlap between keypad, LCD, and P10 interfaces, enabling smooth and conflict-free operation across all modules.

CHAPTER 3

SYSREM DESIGN AND CIRCUIT DESCRIPTION

System Design and Circuit Description

This chapter presents the overall system design of the Token Display System for the Counselling Cell, including the **block diagram**, **circuit explanation**, and **individual module configurations**. The system is developed in a modular fashion where each receptionist acts as an independent unit with input and output interface, while the central controller (Arduino Mega) collects and processes the data to drive the common display (P10 LED matrix).

Block Diagram

The block diagram represents the working principle and data flow of the system. Each of the four receptionist modules consists of:

- A **4x4 Matrix Keypad** for token input
- An **I2C 16x2 LCD** for local display

These modules are connected to a **single Arduino Mega 2560**, which processes the input tokens and updates:

- The LCD screens with confirmation

Refer to Fig. 1: Block Diagram of Token Display

Working Principle:

- The receptionist enters a token number using the keypad.
- The LCD displays the entered token number locally for confirmation.
- The Arduino Mega collects inputs from all receptionist modules.
- It formats and sends the combined data to the P10 LED display.
- The LED matrix shows the tokens in a scrolling or fixed format.

3.2 Circuit Connections

The following explains how the hardware components are connected to the Arduino Mega:

a. Keypad Connections

Each keypad has 8 pins (4 rows + 4 columns). These pins are connected to the digital I/O pins of the Arduino as shown below:

Keypad	Row Pins	Column Pins
Keypad 1	22, 23, 24, 25	26, 27, 28, 29
Keypad 2	30, 31, 32, 33	34, 35, 36, 37
Keypad 3	38, 39, 40, 41	42, 43, 44, 45
Keypad 4	46, 47, 48, 49	50, 51, 52, 53

Each keypad is interfaced using the Keypad.hlibrary and configured independently.

b. LCD (I2C) Connections

All 4 LCDs share the same I2C communication lines:

- **SDA** → Pin A4
- **SCL** → Pin A5
- **VCC & GND** → Common 5V and GND rails

Each LCD is assigned a **unique I2C address** (like 0x27, 0x26, 0x25, 0x24) using the LiquidCrystal_I2C.hlibrary.

c. P10 LED Display Connection

The P10 LED display (3 modules connected in series) is interfaced using SPI pins:

- **DIN (Data In)** → Pin 9
- **CLK (Clock)** → Pin 10
- **CS/STB (Chip Select)** → Pin 11

LCD and Keypad Configuration

Each receptionist module works independently with the following interaction:

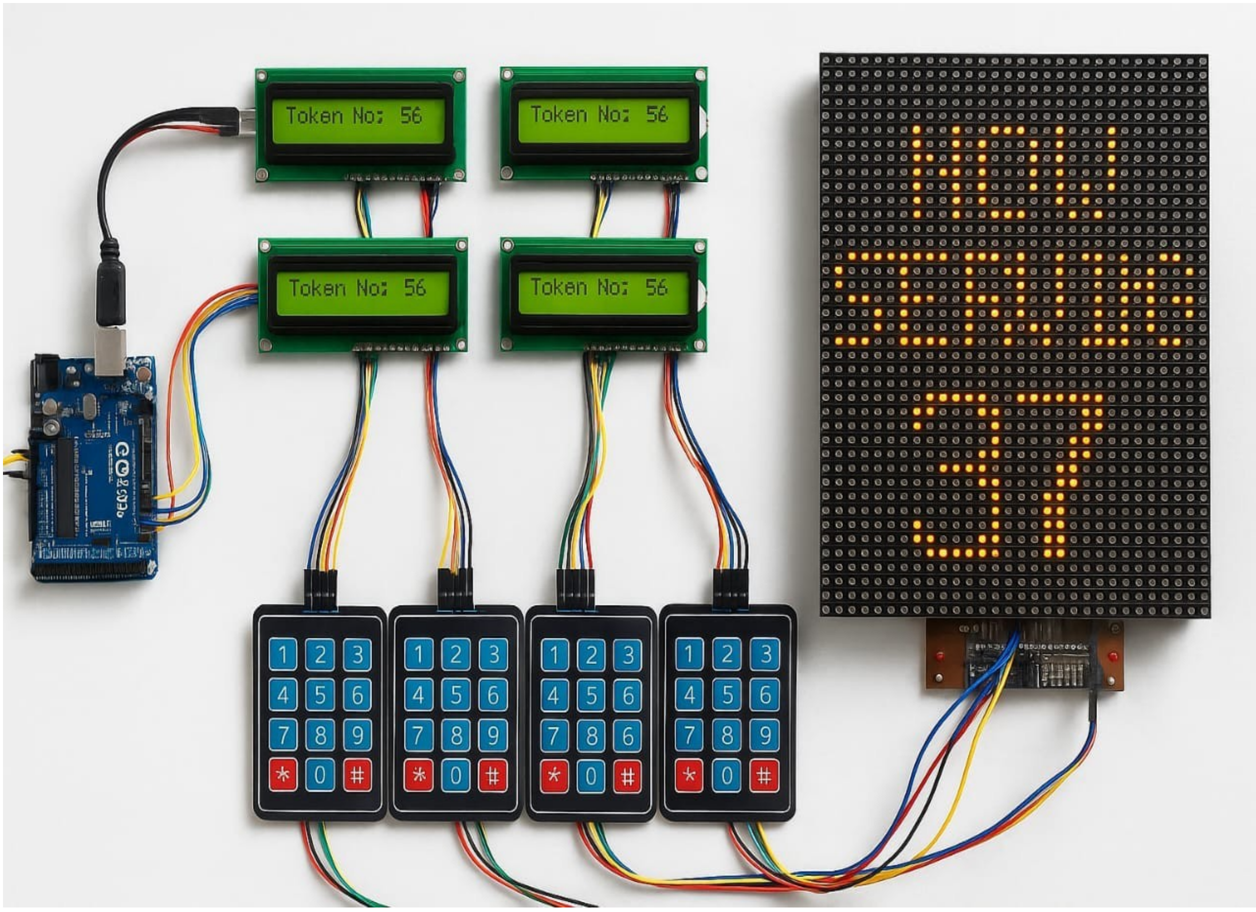
1. **Token Input**: Receptionist types a token number (up to 4 digits) using the keypad.

2. **LCD Display**: The LCD shows “Enter Token” and updates the digits in real-time.

3. **Send Command**: Pressing #submits the token, which:

   - Sends the token to the Arduino
   - Displays confirmation on the LCD (e.g., “Sent: 103”)
   - Updates the scrolling LED display

4. **Clear Command**: Pressing *clears the current input and resets the LCD prompt.



• **Fig. 3.1** Circuit Connections of Keypads to Arduino Mega

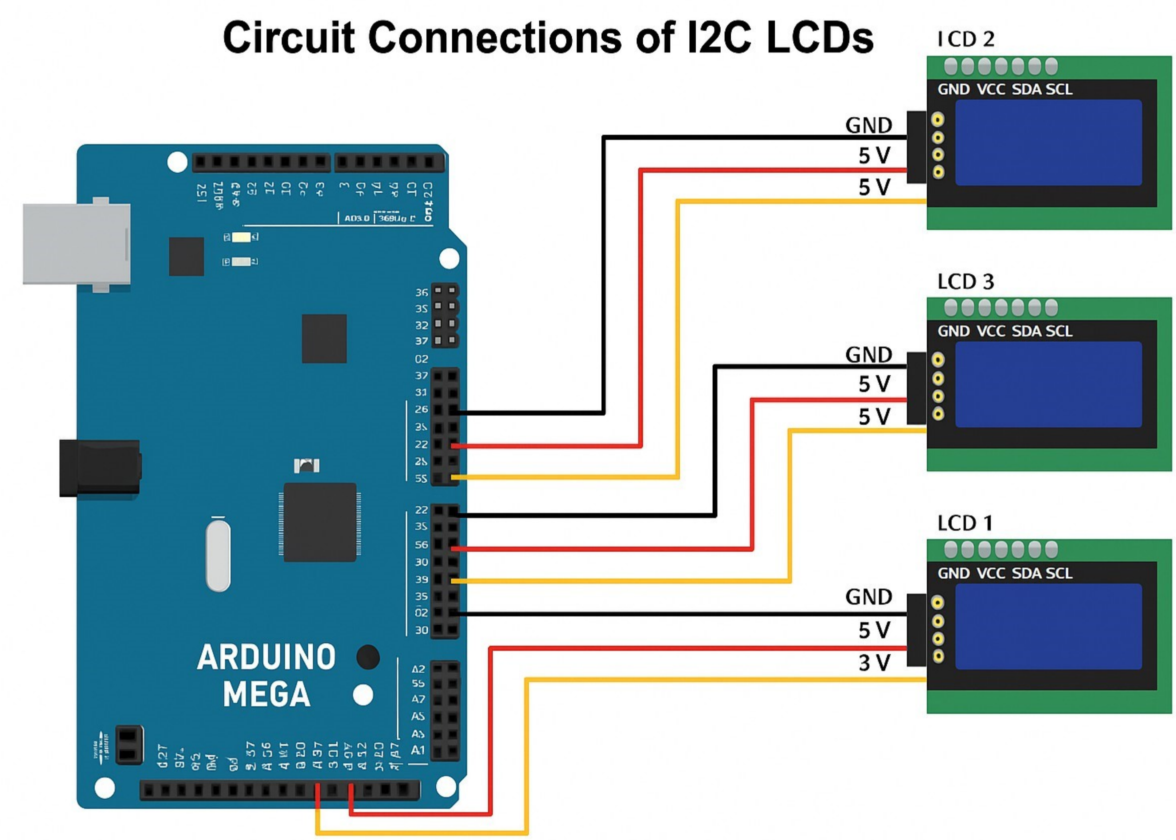


Fig. 3.2: Circuit Connections of I2C LCDs with Arduino Mega

The system uses four **16x2 I2C LCD displays**, one for each receptionist module. These LCDs provide real-time feedback during token entry, confirming each button press and the final token submission. The I2C interface helps in minimising wiring complexity by using only two communication lines: **SDA (Serial Data)** and **SCL (Serial Clock)**.

Since all four LCDs share the same SDA and SCL lines, they are assigned **unique I2C addresses** (e.g., `0x27`, `0x26`, `0x25`, `0x24`) to avoid data collision. The displays are powered through a 5V supply and grounded using a common GND rail.

Each LCD is connected to the Arduino Mega as follows:

SDA → Pin A4

SCL → Pin A5

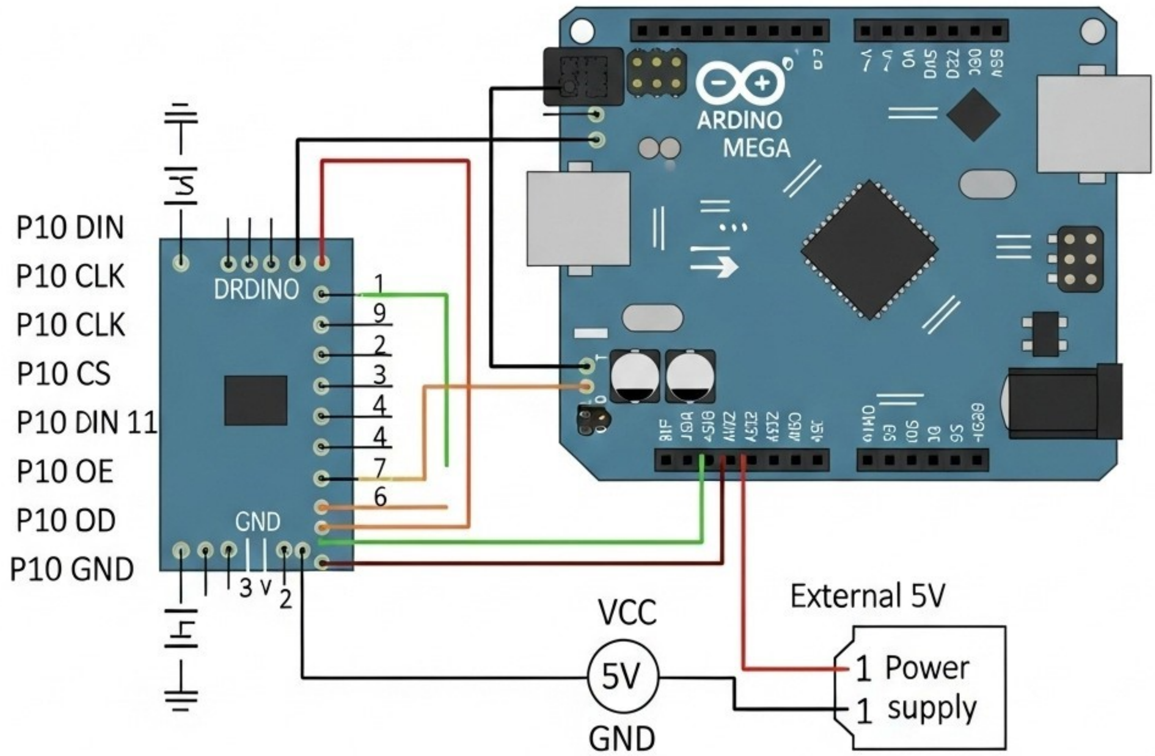
VCC → 5V

GND → GND

The connection setup is shown in the diagram below:

<centre> <img src="insert diagram here" alt="Circuit Connections of I2C LCDs" width="80%"> </centre>

The P10 LED matrix display is connected to the Arduino Mega using SPI- compatible pins: **DIN**, **CLK**, and **CS**. An external 5V power supply is used to meet the current demands of the LED matrix. The wiring is illustrated below.



- Fig3.3.P10 Display

S.NO.	Receptionist Unit	LCD Position	I2C Address
1	Receptionist 1	LCD 1	0*27
2	Receptionist 2	LCD 2	0*26
3	Receptionist 3	LCD 3	0*25
4	Receptionist 4	LCD 4	0*24

- **Table 3.1** LCD I2C Address Mapping

All LCDs are powered using a shared 5V and GND rail and are connected to the Arduino via SDA (A4) and SCL (A5) pins.

CHAPTER 4

CODE EXPLANATION AND LOGIC

Code Explanation and Logic

The functionality of the Token Display System is entirely driven by Arduino- based code. The code is structured into three major parts — setup, input handling, and output control. Each receptionist unit operates independently through a dedicated keypad and LCD, while all token data is displayed together on the central P10 LED display.

The code utilises multiple libraries for effective component handling, including:

Keypad.hfor keypad scanning

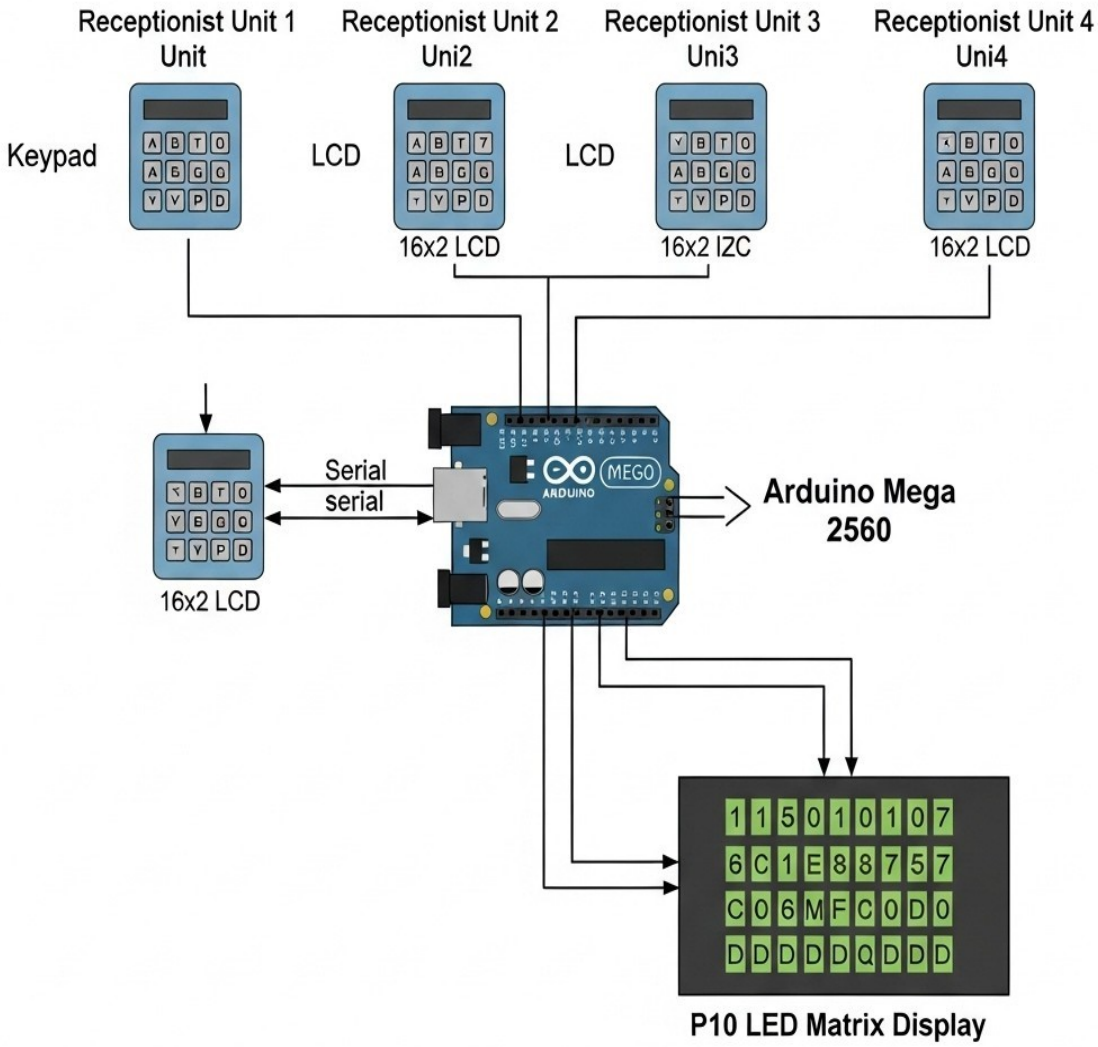
LiquidCrystal_I2C.hfor I2C-based LCD control

MD_Parola.hand MD_MAX72XX.hfor controlling the P10 scrolling LED display

Wire.hand SPI.hfor communication handling

Token Handling Logic

Each keypad is continuously monitored using a function named `handleKeypadInput()`. Whenever a key is pressed, it is captured and displayed on the corresponding LCD screen. When the receptionist presses `#`, the token is confirmed and sent to the P10 display. If `*` is pressed, the current input is cleared.



- fig4.1 Token Flow

The flow diagram below illustrates how each receptionist interacts with the system. Inputs from keypads are processed by Arduino Mega, feedback is shown on LCDs, and final tokens are pushed to the P10 LED display.

🧩de Snippet: Token Input Handling

```
void handle Keypad Input(Keypad &keypad, LiquidCrystal_I2C & lcd, int index) {
char key = keypad. Get Key(); if (key) {
if (key == '#') {
if (tokens[index].length() > 0) { lcd. clear();
lcd. setCursor(0, 0); lcd. print("Sent: " + tokens[index]); tokens[index] = ""; lcd. setCursor(0, 0);
lcd.print("Enter Token:");
}
} else if (key == '*') { tokens[index] = ""; lcd.clear(); lcd.setCursor(0, 0); lcd.print("Enter Token:");
} else {
if (tokens[index].length() < 4) { tokens[index] += key; lcd.setCursor(0, 1); lcd.print(tokens[index]);
}
}
}
}
```

This function ensures that every receptionist has control over their token input, independent of the others.

4.1 Display Control Using P10

All active tokens (those that were submitted using #) are stored in an array and are displayed on the P10 LED board using scrolling text. This is done using the updateP10Display() function. It loops through the token array, constructs a string, and scrolls it across the LED display using the MD_Parolalibrary.

🧩 Code Snippet: P10 Display Update

```
void updateP10Display() {
String displayText = "Tokens: "; for (int i = 0; i < 4; i++) {
if (tokens[i].length() > 0) { displayText += tokens[i] + " ";
}
}

P.displayClear();
P.displayScroll(displayText.c_str(), PA_CENTER, PA_SCROLL_LEFT, 100);
}
```

This function provides real-time scrolling updates on the LED matrix for the waiting area.

Setup and Initialisation

The `setup()` function is used to initialise all four LCDs and the P10 display. Backlight, cursor position, and initial messages are defined here. This function runs once when the Arduino is powered on or reset.

Code Snippet: Initialization

```
void setup() {
 lcd1.begin();  lcd1.backlight();  lcd2.begin();
 lcd2.backlight();                 lcd3.begin();
 lcd3.backlight();                 lcd4.begin();
 lcd4.backlight();

 lcd1.setCursor(0, 0); lcd1.print("Enter Token:"); lcd2.setCursor(0, 0);
 lcd2.print("Enter Token:"); lcd3.setCursor(0, 0); lcd3.print("Enter
 Token:"); lcd4.setCursor(0, 0); lcd4.print("Enter Token:");

 Serial.begin(9600);

 P.begin();
 P.setIntensity(5);
 P.displayClear();
 }
```

Summary

Each receptionist operates an individual unit with keypad and LCD.

Submitted tokens are combined and scrolled on the P10 LED panel.

The system uses efficient code structure and library functions.

- Input and output operations arehandled asynchronously, improving performance.

This code structure ensures scalability, clarity, and modular design — making it easier for further extension like adding new reception units or replacing display types.

Working and Output Demonstration

The Token Display System for Counselling Cell has been designed to simplify and automate token management in institutional setups. The system allows four independent receptionist units to enter token numbers, which are then centrally displayed on a scrolling LED board for public view. The working of the system includes both input from keypads and output on LCDs and P10 LED matrix.

Input via Keypads

Each receptionist is provided with a **4x4 matrix keypad** connected to an **Arduino Mega 2560**.

- The receptionist types the token number using the keypad.
- As digits are pressed, they are displayed live on the associated **16x2 I2C LCD screen**.
- Pressing the #key submits the token, while pressing *clears the current input.

Each receptionist works independently, and their actions do not interfere with the others. This multi-input functionality is achieved using arrays and modular code functions in the program.

Output on LCD and P10

After a token is submitted using the #key:

The **LCD** shows a confirmation message:

➤ *Sent: 105*

Then it resets to the prompt:

➤ *Enter Token:*

The submitted token is also sent to the **central scrolling P10 LED display**, which shows the list of active tokens to the waiting visitors.

The submitted token is also sent to the **central scrolling P10 LED display**, which shows the list of active tokens to the waiting visitors.

- The LED display scrolls messages like:

➤ *Tokens: 105 203 007*

- Display updates are handled using the `MD_Parola` and `MD_MAX72XX` libraries, with dynamic text refreshing and smooth scrolling.

This ensures that visitors are always informed in real time, reducing confusion and increasing service efficiency.

Combined System Flow

- All 4 receptionist units work in parallel.
- Each LCD provides personal feedback to the receptionist.
- Tokens from all receptionists are displayed centrally on the LED board.
- The system is powered by a 5V 10A power supply and controlled by the Arduino Mega.
- No manual syncing is required — the logic and code handle everything in real time.

CHAPTER 5

WORKING AND OUTPUT DEMONSTRATION

Working and Output Demonstration

This chapter presents the overall working of the **Token Display System for Counselling Cell**, focusing on how each module operates and how data is processed and displayed. The system is designed to ensure smooth and real-time communication between multiple receptionist units and a centralised display for visitors.

5.1 Input via Keypads

Each receptionist is equipped with a **4x4 matrix keypad** to enter token numbers. These keypads are connected to the Arduino Mega through separate digital pins, allowing individual operation.

- When a receptionist starts entering a token number, the digits appear in real-time on their respective **I2C 16x2 LCD**.

- Once the token number is complete, the # key is pressed to **submit** the token.

- The * key is used to **clear** the input and re-enter a new token if required.

Each receptionist unit operates independently and does not interfere with the others, allowing multiple token entries simultaneously.

5.2 Output on LCD and P10 Display

After pressing the # key, the token number is sent to two outputs:

a) I2C LCD

- Displays a confirmation message such as:

➤ *Sent: 105*

- Resets to the prompt message:

➤ *Enter Token:*

b)P10 LED Matrix Display

- The Arduino collects all active tokens and **displays them on the P10 scrolling display**.

- The tokens are shown in a continuous scroll as:

➤ *Tokens: 105 204 307*

- The display is updated automatically every time a new token is submitted.

The figure below demonstrates how the Arduino Mega connects to the P10 module and shows the scrolling output.

<center> <img src="insert image here" width="80%"> </center>

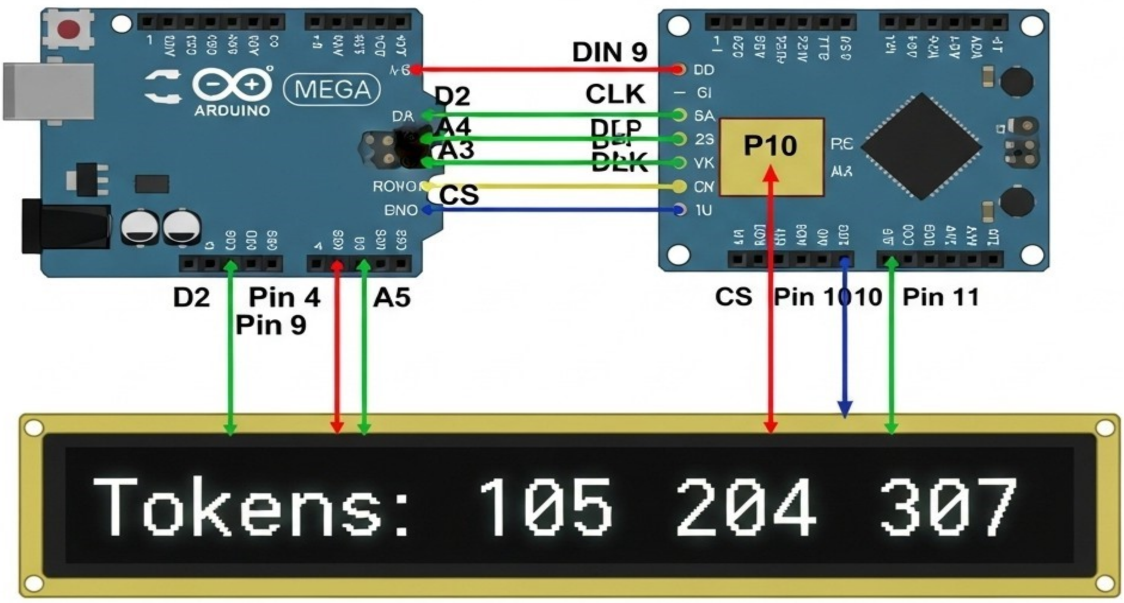


Fig.5.1 Output display on p10

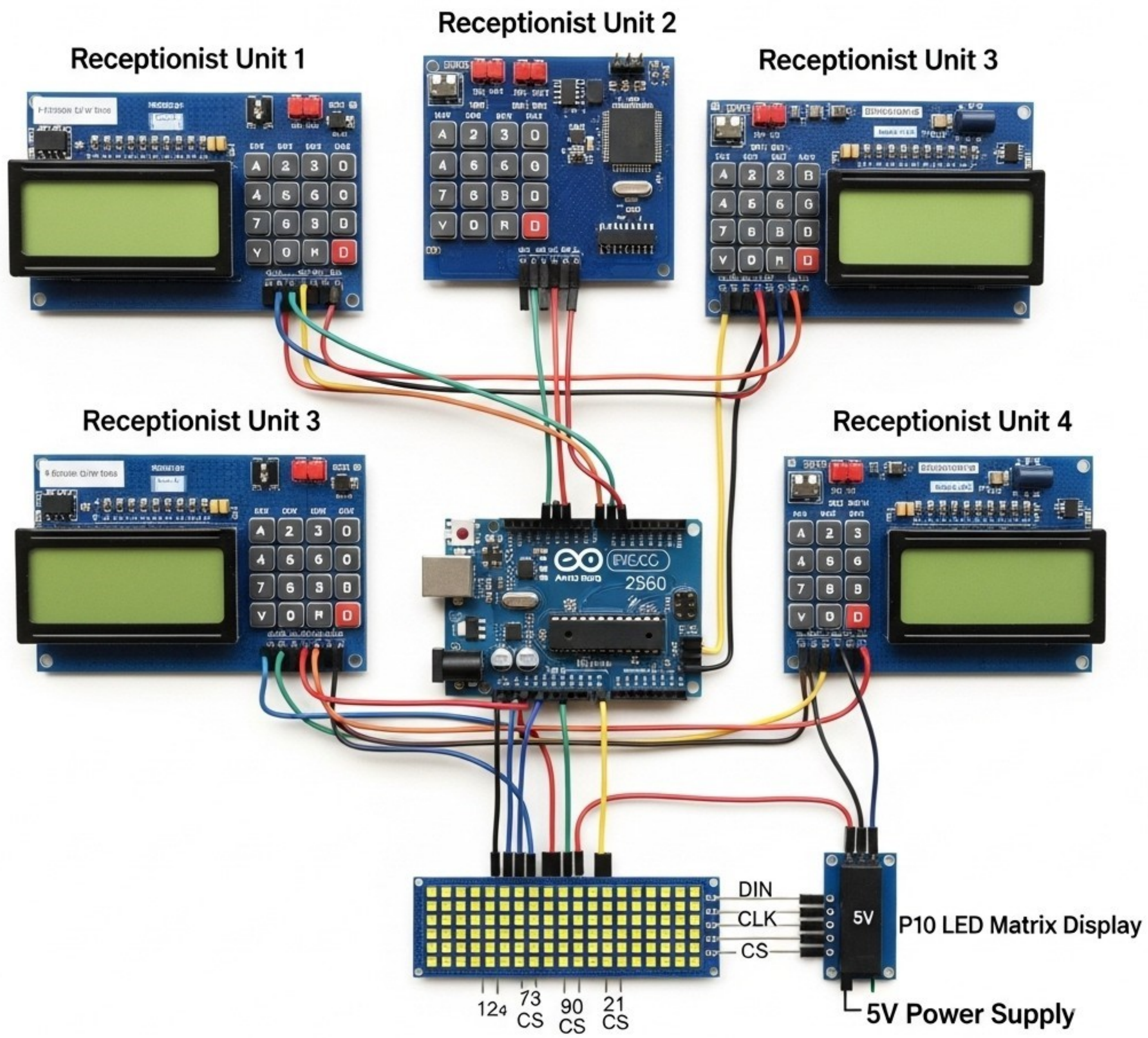


Fig 5.2 final hardware setup of token system

The complete assembled hardware of the token display system is shown below. It includes four receptionist modules with keypad and LCD, all connected to the central Arduino Mega controller. The token outputs are displayed via a P10 LED matrix powered by a 5V supply.

CHAPTER 6

APPLICATION AND FUTURE ENHANCEMENTS

Applications and Future Enhancements

The Token Display System developed in this project has been designed for practical real-world use in institutions where managing queues and organising turn-based services is essential. The system is scalable, reliable, and easy to operate. This chapter highlights both the current applications and potential future improvements.

Real-World Applications

The system can be readily implemented in several environments where crowd management and service scheduling are required. Some common applications include:

Educational Institutions

Counselling Cells: To manage student appointments efficiently across multiple desks.

Library Counters: For tracking issue/return requests.

Healthcare Facilities

Hospitals & Clinics: To handle patient queues in OPDs, labs, and reception areas.

Banks and Government Offices

Token Management: At enquiry counters, bill payment desks, and help desks.

Private Offices & Service Centres

Customer Service: To handle technical support or customer walk-ins in an orderly way.

6.1 Future Scope

While the current system performs well within its scope, several enhancements can make it more advanced and adaptable to modern digital needs.

1. Wireless Token Entry

- Using Wi-Fi-enabled modules (like ESP8266 or NodeMCU), token input can be done wirelessly from mobile apps or browser interfaces.

2. Voice Announcement System

- Integration with a speaker module to announce token numbers aloud along with display.

3. Cloud-Based Monitoring

- Store token logs online for real-time access by administrators or for analytics.

4. Mobile App Integration

- Create a companion app for receptionist-side token entry and user-side token status checking.

5. AI-Based Load Management

- Smart token assignment based on queue lengths or service time can be added using basic logic or AI algorithms.

CHAPTER 7

CONCLUSION

Conclusion

The Token Display System for Counselling Cell, as developed and implemented in this project, offers a practical and efficient solution for managing queues in institutional environments. The system was designed to address common issues such as crowd mismanagement, delayed services, and poor communication between visitors and service providers.

By integrating multiple receptionist input modules through keypads and LCDs with a centralised scrolling LED display, the project successfully demonstrated real-time, multi-point token management. The use of the Arduino Mega 2560 microcontroller provided the flexibility needed to interface with multiple peripherals while maintaining simplicity in design.

The modular architecture of both hardware and software ensured that each receptionist unit operated independently without conflict, while the P10 LED matrix display provided a synchronised and public-friendly output. The system's design is easily extendable and allows for future enhancements such as voice announcements, mobile app integration, and cloud-based monitoring.

This project not only fulfils its objective of improving service efficiency in counselling cells but also serves as a foundational example of how embedded systems can be applied to solve real-world problems in queue management and public service systems.

CHAPTER 8

REFRENCES

References

Below is the list of books, websites, libraries, and tools referred to during the design and development of the **Token Display System for Counselling Cell**:

1. **Arduino Official Documentation**
https://www.arduino.cc
→ For understanding the Arduino Mega 2560 and IDE setup.

2. **MD_Parola and MD_MAX72XX Library Documentation**
https://majicdesigns.github.io
→ For implementing scrolling text on the P10 LED display.

3. **Keypad and LiquidCrystal_I2C Arduino Libraries**
Arduino Playground and GitHub repositories
→ Used for interfacing the 4x4 matrix keypad and 16x2 LCD displays.

4. **Datasheets**

◦ P10 LED Module (HUB75)

◦ Arduino Mega 2560 Pinout

◦ I2C LCD Module

◦ 4x4 Matrix Keypad

5. **Electronics Hub – Embedded Systems Projects**
https://www.electronicshub.org

6. **TutorialsPoint – Arduino Basics**
https://www.tutorialspoint.com/arduino

7. **YouTube Channels:**

◦ "Techiesms" – Arduino P10 Display Tutorial

◦ "Indian LifeHacker" – LCD and Keypad Interfacing
→ For visual references and testing ideas

◦

Abbreviation	Full Form
LCD	Liquid Crystal Display
P10	32x16 LED Matrix Display
I2C	Inter-Integrated Circuit (Communication Protocol)
VCC	Voltage at the Common Collector (+5V)
GND	Ground
MCU	Microcontroller Unit
IoT	Internet of Things
SDA	Serial Data Line (I2C)
SCL	Serial Clock Line (I2C)
CLK	Clock (P10)
STB/CS	Strobe / Chip Select (P10)
OE	Output Enable
DIN	Data In (P10)